HUY HUNG NGUYEN

Auburn, Alabama
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EDUCATION & ACHIEVEMENTS

Ph.D - Auburn University 2023 - Now

Computer Science and Software Engineering

Master - Seoul National University 2015 - 2018

Electrical and Computer Engineering
Thesis Title: Bit Transpose Unit Design for Approximate Memory Architecture
GPA: 3.8/4.0

Bachelor - Ha Noi University of Science and Technology 2007 - 2012

Automatic Control (Honor Class)
Thesis Title: Novel Design for Barrier Lyapunov Function
GPA: 3.2/4.0

WORK EXPERIENCE

Atomi Digital January 2021 - August 2023

Career Position: R&D Manager
Deep Learning and Computer Vision
- Face Verification, Face Identification, and Face-Anti Spoofing (a component in the eKYC system)
- OCR for Identity Card (a component in the eKYC system)
- Smart Queue in Smart Banking Branch (detect, track customer, and insert in the queue)
- Our eKYC system served over two million customers of Sacombank and Lien Viet Post Bank

Big Data
- Design and implement a system for storing and analyzing data on banking customer behavior

System Architecture
- Upgrade the banking systems in the micro-services architecture using K8S and OpenShift

Asilla Viet Nam December 2018 - January 2021

Career Position: AI R&D Team Leader
Deep Learning and Computer Vision
- 2D human pose estimation and tracking
- 3D human pose estimation
- Unsupervised/Semi-supervised abnormal detection
- Multiple camera tracking
- Compress human pose deep learning model using knowledge distillation
- Object detection and tracking on sports applications (basketball and volleyball)
- Design, implement, and optimize new deep learning models on ARM for IoT applications

Viettel Research and Development Institute - Viettel Group June 2012 - February 2015

Career Position: Automation Engineer
Design the Electric Automatic Systems on Radar Vehicles
- Design electric systems for the Radar systems
- Deploy automatic algorithm in Siemens Programmable Logic Controller (PLC) devices, SCADA network
- Upgrade the automatic systems in many Radar centers

PUBLICATIONS

TAB: Transformer Attention Bottlenecks enable User Intervention and Debugging in Vision-Language Models
Pooyan Rahmanzadehgervi, Hung Huy Nguyen, Rosanne Liu, Long Mai, Anh Totti Nguyen
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025, pp. 22551-22562

Improving Zero-Shot Object-Level Change Detection by Incorporating Visual Correspondence
Hung Huy Nguyen, Pooyan Rahmanzadehgervi, Long Mai, Anh Totti Nguyen
2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

An Approximate Memory Architecture for Energy Saving in Deep Learning Application
Duy Thanh Nguyen, Nguyen Huy Hung, Hyun Kim, and Hyuk-Jae Lee
IEEE Transactions on Circuits and Systems I: Regular Papers Volume: 67, Issue: 5, Pages: 1588-1601, May. 2020

Bit Transpose Unit Design for Approximate Memory Architecture
Nguyen Huy Hung, Duy Thanh Nguyen, and Hyun Kim
2018 Summer Conference

Parallel Execution of CT Image Reconstruction with a GPU
Nguyen Huy Hung, Hyuk-Jae Lee, Dong-Goo Kang, and Seung-Oh Jin
32nd International Technical Conference on Circuits/Systems, Computers and Communications

Three-Dimensional Facial Reconstruction Using Two-Dimensional Images Captured from Cellphones
Nguyen Huy Hung, Sunwoong Kim, and Hyuk-Jae Lee
2016 Summer Conference

PATENT

AI Smart Warning. Patent number: 5308/2020/QTG, Viet Nam

HONORS & AWARDS

Auburn University Presidential Graduate Research Fellowship 2023-2026
Seoul National University ECE Department Scholarship 2016-2017
Seoul National University Global Scholarship 2015-2016

PROGRAMMING SKILLS

C/C++	Professional
PyTorch, Keras	Professional
Python, Matlab	Professional
OpenCV, OpenGL, CUDA	Advanced
PLC, SolidWork, AutoCad	Advanced

REFERENCES

Prof. Anh Totti Nguyen, Auburn University, AL, U.S. (anhnguyen@auburn.edu)
Prof. Hyuk-Jae Lee, Seoul National University, Seoul, S. Korea (hyukjae@snu.ac.kr)